

13 C $\frac{I_1}{I} = \cos^2 20^\circ$, and $\frac{I_2}{I_1} = \cos^2 (\theta - 20^\circ)$

$$\therefore \frac{I_1}{I} \times \frac{I_2}{I_1} = \frac{I_2}{I} = \cos^2 20^\circ \cos^2 (\theta - 20^\circ) = 0.25$$

$$\therefore \cos 20^\circ \cos (\theta - 20^\circ) = 0.50$$

$$\therefore \cos (\theta - 20^\circ) = \frac{0.50}{\cos 20^\circ} = 0.53209$$

$$\therefore \theta - 20^\circ = 57.853^\circ \Rightarrow \theta = 77.853^\circ \approx 78^\circ$$

14 D A stationary wave forms between C and H. As such the phase difference between E and F